Satellite Design Recommendation

Mission #24 • Generated from a saved analysis

Mission brief

Monitor vegetation health and irrigation conditions across farms near Taif every five days using multispectral imagery.

Conceptual design

Design field	Recommendation
Mission type	Earth Observation
Orbit	SSO
Altitude	600 km
Payload	Multispectral Imager
Power	500 W
Mass class	Small
Lifetime	5 years
ADCS	Three-axis stabilized

Engineering justification

Sun-synchronous orbit (SSO) at approximately 600 km altitude allows consistent lighting conditions for multispectral imaging, crucial for monitoring vegetation health. Revisiting the area every five days is achievable with a single satellite or small constellation in SSO. A multispectral imager provides the necessary spectral bands for detailed vegetation and irrigation analysis. A small satellite class with about 500 W power supports payload and communication needs. Three-axis stabilization ensures precise pointing and image quality. A 5-year lifetime balances cost and mission requirements for continuous monitoring.

Engineering knowledge

Design drivers

- Spatial resolution
- Revisit time
- Swath width
- Data volume
- Lighting conditions

Required subsystems

- ADCS - Accurate pointing is required for stable and correctly targeted imagery.
- OBC - Image data may require onboard processing, compression, and storage.
- COMMS - Large payload data volumes require a suitable downlink.
- EPS - The payload, processor, and communication system require reliable power.

Payload options

- RGB Camera - Natural-color imagery is required for mapping or visual inspection.
- Multispectral Camera - Vegetation, agriculture, water, or environmental analysis is required.
- Synthetic Aperture Radar - Nighttime or all-weather observation is required.

Advantages

- SSO provides approximately consistent lighting conditions.
- LEO provides better spatial resolution than higher-altitude orbits.

Limitations

- A single satellite does not provide continuous global coverage.
- High-resolution imagery produces large data volumes.

Selection rules

- The mission focuses on agriculture or vegetation. - Use a multispectral payload in SSO.
- The mission requires nighttime or cloud-penetrating observation. - Use a SAR payload in LEO or SSO.
- The mission requires simple visual mapping. - Use an RGB camera in LEO or SSO.